

VASILEIOS KARAGIANNIDIS

Junior Software / Embedded Engineer · Robotics Researcher

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student with hands-on experience in Python, embedded systems (ESP32, Raspberry Pi) and research work in computer vision and mobile manipulation with the Boston Dynamics Spot robot on NVIDIA Isaac Sim. Vice President of the IEEE IHU Student Branch and co-organizer of the IEEEcon 2026 conference. I combine five years of leadership in a high-pressure environment with strong technical specialization.

Looking for: Internship or Junior position in Software Development, Embedded Systems, Robotics or IoT teams — full-time, on-site/hybrid in Thessaloniki or remote.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL, C/C++ (embedded)

Python Stack: yt-dlp, Tkinter (GUI), smtplib/IMAP, automation scripts

Embedded & Hardware: ESP32, Raspberry Pi, microcontrollers, circuit design & analysis

Robotics & Simulation: NVIDIA Isaac Sim, Boston Dynamics Spot SDK, Computer Vision, Mobile Manipulation

Tools & Platforms: Git, GitHub, Linux, VS Code

Areas of Interest: Robotics, IoT, Embedded Systems, Computer Vision, Automation

PROFESSIONAL & ACADEMIC EXPERIENCE

Researcher — Robotic Simulation & Vision — International Hellenic University 2025 – Present

- Set up and developed a simulation environment in NVIDIA Isaac Sim for the Boston Dynamics Spot quadruped robot
- Working on the Spot vision system — extracting and processing visual data from the robot's sensors
- Implementing mobile manipulation tasks: autonomous navigation to a target and object handling via the robotic arm (pick & place)
- Continuous deepening in the simulation environment, configuration and robotic application development workflow

Vice President — IEEE IHU Student Branch 2025 – 2026

- Co-organizer of the two-day IEEEcon 2026 conference — inviting and coordinating speakers, liaising with companies and academic teams, securing sponsorships
- Co-organized the branch's participation in the international IEEE Extreme competition (October 2025) — secured sponsorships and achieved a high final ranking
- Designed, organized and delivered technical workshops for students: 'Introduction to Microcontrollers' and 'Microcontrollers & Robotics'
- Developed an in-house email automation tool for IEEEcon communications (see automailforieee project)

Kitchen Coach — KFC Thessaloniki 2020 – 2025

- Training and mentoring of new kitchen staff — onboarding & coaching
- Kitchen team supervisor and shift workflow management
- Mediator for store-level operational issue resolution
- Enforcement of Food Safety standards

SELECTED PROJECTS

Security-System-with-ESP32 — ESP32-based security system — sensors, real-time alerts and control.

Tech: ESP32, C/C++, sensors, embedded

automailforieee — Automated email sending system for the IEEEetcon conference — communication with speakers, sponsors and attendees.

Tech: Python, smtplib, email automation

automailtech — Automated email sender targeting a list of tech companies — cold outreach tool for partnerships & sponsorships.

Tech: Python, CSV processing, email automation

youtube-to-mp3 — Desktop app for quick YouTube-to-MP3 conversion with a clean, user-friendly interface.

Tech: Python, yt-dlp, Tkinter (GUI)

movie-planner — Online real-time watchlist for movies — personal organization tool.

Tech: Python

Fast-Food-App — Order management application for a fast-food store — inspired by my KFC experience.

Tech: —

ItsMyBirthday — Personal birthday invitation website — interactive front-end project.

Tech: JavaScript, HTML, CSS

All projects: github.com/karagiancode

EDUCATION

International Hellenic University (IHU) — Thessaloniki

2019 – Present

Department of Information & Electronic Systems Engineering — Specialization: Computer Science & Programming

30th General Lyceum of Thessaloniki

2016 – 2019

High School Diploma

CERTIFICATIONS & LANGUAGES

- International Diploma in IT Skills — Cambridge
- Basic Life Support (BLS) — European Resuscitation Council (ERC)

Languages: Greek (native) · English (B2 — good command)

ADDITIONAL INFORMATION

- Upcoming volunteer work with the TechSaloniki organization (2026) — contributing to the local tech community
- Soft skills: Leadership, teamwork, working under pressure, communication, eager to learn and adapt
- Interests: Tae Kwon Do (WTF, 2013 – 2018), Chess, Gymnastics